# Capstone Project 3 Proposal

**Project Title:** The K-Beauty Spa Architect: A Multi-Tiered AI Personal Assistant for One-stop Skincare

**Domain:** Health, Wellness, & Retail AI (K-Beauty Focus)

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## 1. The Problem Statement

Despite the popularity of K-Beauty, consumers face high "choice paralysis" due to complex ingredient lists (e.g., Snail Mucin, Mugwort, Centella) and inconsistent availability across retailers like Sephora and Costco. Friends and spa clients need a data-driven "Skin Prescription" that balances clinical efficacy with budget constraints, while also providing a roadmap for professional spa procedures like Botox or microneedling.

## 2. Project Goal

To develop an intelligent recommendation engine that:

1. **Extracts Intent:** Processes vague user queries (e.g., "sun stains on dark skin") using NLP.
2. **Generates Routines:** Constructs a 4-step daily regimen (Cleanse, Treat, Moisturize, Protect).
3. **Applies Budget Tiers:** Offers comparative options across **Costco Value**, **Optimal K-Beauty**, and **Luxury Clinical** tiers.
4. **Integrates Spa Logic:** Recommends professional procedures that complement the recommended topicals.

## 3. Data Strategy & System Architecture

### I. Core Datasets

- **Primary Product Data:** Sephora `product_info.csv` (1,000+ skincare products).
- **Curated Knowledge Base:** A custom "Brand-Tier" mapping (e.g., COSRX = Costco Value; Sulwhasoo = Luxury).
- **Ingredient Dictionary:** A cross-reference table mapping K-Beauty botanicals to clinical functions (e.g., Hanbang -> Antioxidant).

**II. The 3-System Implementation (Technical Modules)**

1. **NLP Module (Intent & Semantic Search):** * Uses **Sentence-Transformers** to map human concerns to a vector space.
    - Identifies "Hero Ingredients" required for the specific skin type (Acne, Hyperpigmentation, Sensitivity).
2. **Recommendation System (Tiered Logic Engine):**
    - Calculates a **"Suitability Score"** for each product based on ratings, reviews, and ingredient match.
    - Groups products by price point to create the 6-month budget roadmap.
3. **Integrative Logic (Spa Advisor):**
    - A rule-based expert system that triggers clinical advice (e.g., Botox for forehead lines) based on identified concerns.

**The "Commercial Tiers" Strategy**

- **Tier 1: Costco Budget:** Focuses on bulk-available, high-value K-Beauty/Clinical brands (e.g., COSRX, Dr. Jart+).
- **Tier 2: Medium:** Accessible luxury brands available at Sephora (e.g., Laneige, Innisfree).
- **Tier 3: Optimal (K-Beauty Professional):** High-efficacy brands often used in Korean clinics (e.g., Sulwhasoo, Beauty of Joseon).
- **Tier 4: Luxury Clinical:** Premium brands for post-procedure care (e.g., Tatcha, SK-II).

## 4. Pre-Generated Brand Tier & Ingredient Mapping

To save you research time, here is the "Knowledge Base" for your code:

| Tier | Brand Focus | Key Logic |
|---|---|---|
| **Costco / Budget** | COSRX, Innisfree, Dr. Jart+, Laneige | High efficiency, low price-per-oz. Often available in bulk. |
| **Medium / Optimal** | Beauty of Joseon, Glow Recipe, Belif | Trendy, high social engagement, specialized in K-Beauty "Glass Skin." |

| Tier | Brand Focus | Key Logic |
|---|---|---|
| **Luxury / Clinical** | Sulwhasoo, Tatcha, SK-II, SkinCeuticals | High price point, "Hanbang" ingredients, prestige branding. |

**Skin-Type Hero Ingredient Map:**

- **Hyperpigmentation/Dark Skin:** Arbutin, Tranexamic Acid, Niacinamide, Vitamin C.
- **Acne/Blackheads:** Salicylic Acid (BHA), Tea Tree, Centella (for healing).
- **Wrinkles (Botox Support):** Acetyl Hexapeptide-8 (Argireline), Copper Peptides, Retinol.
- **Sensitive/Stains:** Mugwort, Rice Water, Ceramides.

## 5. Success Metrics & Future Extensions

- **Technical Success:** Measured by the precision of the semantic search (retrieving relevant products for vague queries).
- **Commercial Success:** A clear, printable "Prescription Card" for spa clients.

### Options for expansion:

**A) Recommendation System (Tiered Logic):**

- **Logic:** A custom multi-label filter that buckets products by price percentile and brand reputation.
- **Scheduling:** A temporal logic layer that calculates **Product Life Cycles**. (e.g., A 30ml serum used twice daily lasts ~45 days; the system schedules your next "Costco Run" or "Spa Visit" accordingly).

2. **B)** Integrating Deep Learning CV & **Image Processing** in Phase 2 to allow users to upload a selfie for automated skin analysis, for next version.
   - **Data:** SCIN (Skin Condition Image Network) public dataset.
   - *Image of a deep learning model for skin condition classification*
   - **Model:** A pre-trained **ResNet-50** CNN to identify skin concerns (Acne, Redness, Hyperpigmentation) from a photo to automate the input layer.